

Implementation

1. Create Views

Query 1.1: Create Student-Course View

View joining Student, Enrollment, and Course tables

```
mysql> CREATE VIEW Student_Course_View AS
-> SELECT
->     s.StudentID, s.Name AS StudentName, s.Email,
->     c.CourseID, c.CourseName, c.Credits
-> FROM Student s
-> JOIN Enrollment e ON s.StudentID = e.StudentID
-> JOIN Course c     ON e.CourseID = c.CourseID;
```

Query OK, 0 rows affected (0.01 sec)

Query 1.2: Create Course-Instructor View

View joining Course and Instructor tables

```
mysql> CREATE VIEW Course_Instructor_View AS
-> SELECT
->     c.CourseID, c.CourseName,
->     i.Name AS InstructorName, i.Department
-> FROM Course c
-> JOIN Instructor i ON c.InstructorID = i.InstructorID;
```

Query OK, 0 rows affected (0.00 sec)

2. Display View Data

Query 2.1: Select All from Student_Course_View

```
mysql> SELECT * FROM Student_Course_View;
```

StudentID	StudentName	Email	CourseID	CourseName	Credits
1	Ramesh	ramesh@gmail.com	201	DBMS	4
3	Sneha	sneha@gmail.com	201	DBMS	4
2	Amit	amit@gmail.com	202	Data Science	3

3 rows in set (0.00 sec)

Query 2.2: Select All from Course_Instructor_View

```
mysql> SELECT * FROM Course_Instructor_View;
```

```
+-----+-----+-----+-----+
| CourseID | CourseName | InstructorName | Department |
+-----+-----+-----+-----+
|      201 | DBMS      | Dr. Patil      | IT         |
|      202 | Data Science | Dr. Sharma     | CS         |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

3. Update Data Using View

Query 3.1: Update Student Name via Student_Course_View

```
mysql> UPDATE Student_Course_View
-> SET StudentName = 'Ramesh Patil'
-> WHERE StudentID = 1;
```

```
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

4. Delete Data Using View

Query 4.1: Attempt to Delete from Join View

```
mysql> DELETE FROM Student_Course_View
-> WHERE StudentID = 2;
```

```
ERROR 1395 (HY000): Can not delete from join view 'StudentDB.Student_Course_View'
```

5. Create Indexes

Query 5.1: Create Index on Student(Name)

```
mysql> CREATE INDEX idx_student_name ON Student(Name);
```

```
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Query 5.2: Create Index on Course(CourseName)

```
mysql> CREATE INDEX idx_course_name ON Course(CourseName);
```

```
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Query 5.3: Create Index on Enrollment(StudentID)

```
mysql> CREATE INDEX idx_enrollment_student ON Enrollment(StudentID);
```

```
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

6. Query Performance Analysis (EXPLAIN)

Query 6.1: EXPLAIN — Search Student by Name

```
mysql> EXPLAIN SELECT * FROM Student WHERE Name = 'Ramesh Patil';
```

```
+----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key |
| key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Student | NULL | ref | idx_student_name | idx_student_name |
| 203 | const | 1 | 100.00 | NULL |
+----+-----+-----+-----+-----+-----+
1 row in set, 1 warning (0.00 sec)
```

Query 6.2: EXPLAIN — Search Course by Name

```
mysql> EXPLAIN SELECT * FROM Course WHERE CourseName = 'DBMS';
```

```
+----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key |
| key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE | Course | NULL | ref | idx_course_name | idx_course_name |
| 203 | const | 1 | 100.00 | NULL |
+----+-----+-----+-----+-----+-----+
1 row in set, 1 warning (0.00 sec)
```

7. Search Queries (Optimized with Index)

Query 7.1: Search Student by Name

```
mysql> SELECT * FROM Student WHERE Name = 'Ramesh Patil';
```

```
+-----+-----+-----+-----+-----+
| StudentID | Name          | Email              | Age | Address |
+-----+-----+-----+-----+-----+
|          1 | Ramesh Patil | ramesh@gmail.com   | 20  | Pune    |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Query 7.2: Search Course by Name

```
mysql> SELECT * FROM Course WHERE CourseName = 'DBMS';
```

```
+-----+-----+-----+-----+
| CourseID | CourseName | Credits | InstructorID |
+-----+-----+-----+-----+
|        201 | DBMS       |        4 |          101 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```